

Code-Layout, Readability and Reusability

Project: Emotion Detection & Learning Support Engine

Date: 25 June 2026 **Team ID:** XXXXXX **Marks:** 5

Quality assessment of the Emotion Detection & Learning Support Engine codebase.

Code Layout Checklist

S.No	Parameter	Description	Followed	Remarks
1	Indentation	Uniform 4-space indentation	Yes	PEP8
2	File Structure	Modular Python project	Yes	Well organized
3	Variables	Meaningful variable names	Yes	Readable
4	Functions	Descriptive function names	Yes	Standard
5	Comments	Inline comments	Yes	Documented
6	Modular Design	Separate ML/API/UI modules	Yes	Maintainable
7	No Redundant Code	Optimized implementation	Yes	Clean
8	Error Handling	Graceful exception handling	Yes	Fallbacks

Reusable Components

Module	Technology	Reuse	Level
Emotion Detector	Scikit-learn	Prediction	High
Preprocessing	Python	Cleaning	High
Gemini API	Requests	AI Response	High
Dashboard	Plotly	Analytics	Medium
Streamlit UI	Streamlit	Frontend	High

Overall Code Quality Assessment

Aspect	Rating	Comments
Structure	5	Excellent modular architecture
Readability	5	Easy to understand
Reusability	5	Reusable modules
Documentation	5	Well commented
Overall	5	Ready for submission